Math for CS & AI: Homework 11

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(Adapted from earlier versions of this course taught by Andrew Yao)

Due Date: Jun 7, 2026

Study Tips

Some students have asked for additional exercises to practice their problem-solving skills. These exercises are not required, but if you are interested, you may find some useful problems in the following textbook:

- *Foundations of Data Science*, by Avrim Blum, John Hopcroft, and Ravindran Kannan
  - Chapter 2: High-Dimensional Space
- *All of Statistics: A Concise Course in Statistical Inference*, by Larry Wasserman
  - Chapter 5: Convergence of Random Variables
  - Chapter 6: Models, Statistical Inference and Learning

You may also want to practice your calculus skills to truly master continuous probability. Check any standard calculus textbook for exercises on integration and differentiation.

You may also want to practice your linear algebra skills, especially those for symmetric matrices. Check any standard linear algebra textbook for exercises. Personally, my favorite linear algebra textbook is:

- *Linear Algebra Done Right*, by Sheldon Axler

Note: Skip the problems that require more knowledge than what is covered in this course.

Problem Set

- Exercises 5.19 from Chapter 5, 6.1–6.3 from Chapter 6, and 7.1–7.2 from Chapter 7 of the lecture notes:

  https://cloud.tsinghua.edu.cn/d/57ddec714cf74c1cab85/

  Six problems in total! Don't miss any of them.

Writing Guidelines

- Same as the first homework.
- We may deduct up to 3 points for poor writing.
- **Special Note:** Your submission should contain only a single PDF file. Please do not submit other files such as `*.zip` or `*.tex`.

Additional Hints

- We provide additional hints for some of the homework problems in the next few pages.
- If you get stuck on a problem, scroll down to find that problem and read the hints.
- Scroll *very slowly* so that you do not accidentally see hints for problems you have not thought about yet.
- Homework is meant to help you practice your problem-solving skills. It is your job to make sure you have thought enough before reading the hints. Ultimately, there will be no additional hints and no AI tools in the exams.
- Again, **peer learning is strongly encouraged**. Reading hints or getting help from AIs is not the same as learning from your classmate who has thought about the problem and can explain it to you. Make some friends who are willing to discuss problems or even have hotpot with you. Learn from their way of thinking, including both successful and unsuccessful attempts.

The rest of this document contains hints for the homework problems.



Think twice before scrolling down!

Hints

Problem 5.19

- The inner product of a random vector with a Gaussian vector follows a symmetric distribution.
- If you get stuck, it may be because you are not yet fully comfortable with linear algebra. In that case, try writing everything in terms of scalar variables first.

Problem 6.1

- Use the PDF criterion.

Problem 6.2

- Try writing everything as sums and sample averages.
- Recall the law of large numbers.
- Recall the central limit theorem.

Problem 6.3

- Use the CDF criterion.
- The approximation formula for $\bar{\Phi}(x)$ may be helpful.
- If there is a $\log n$ in your answer, you are probably on the right track.

Problem 7.1

- This one is easy. Read the lecture notes!
- $\rho(\eta)$ is increasing in η when $\eta > \eta^*$, and decreasing in η when $\eta < \eta^*$.

Problem 7.2

- Read the lecture notes! The idea is almost the same as the positive definite case.
- If $\lambda_i < 0$, then $(1 - \eta\lambda_i) > 1$.
- You just need to make sure that $\alpha_{0,i} \neq 0$.
